

Dynamic fragmentation under laser driven shock loadings

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The fragments ejected from metallic shells subjected to intense laser irradiation are a major concern for the design of the experiments to be performed in large scale laser facilities such as the National Ignition Facility in the USA or the Laser MégaJoule in France. More generally, the ability to anticipate the debris resulting from shock-induced fragmentation and their ballistic properties is a crucial safety issue in many fields, including pyrotechnics. Depending on surface roughness, sample thickness and shock pressure, the main processes governing fragment generation are microjetting (production of $\sim\mu\text{m}$ -scale ejecta upon shock breakout from geometrical defects such as pits or grooves), spall fracture in solid state, microspalling in (partially or fully) liquid state after shock-induced melting, and dynamic punching. Laser driven shocks can be used to investigate these processes over ranges of very high strain rates ($\sim 10^7 \text{ s}^{-1}$), high loading pressures, small spatial scales and very short durations of pressure application (ns-order). Besides, they allow easier sample recovery than the more conventional shock loading techniques based on explosives or plate impacts. We will present an overview of recent experimental work on microjetting, spall fracture and microspall, involving complementary techniques including fast transverse shadowgraphy, time-resolved velocimetry, ultrafast in-situ radiography, and post-recovery analyses of both sample and ejecta. Related modelling efforts will be illustrated more briefly.